Grade: Check Plus

Assignment 2 (Dhaval Patel / dp129@stern.nyu.edu)

(1) Compute the mean and standard deviation of the stock price.

Stock Price (x)	Probability f(x)	xf(x)	$(x-25.41)^2$ f(x)
20	0.05	1.00	1.46
21	0.06	1.26	1.17
22	0.07	1.54	0.81
23	0.08	1.84	0.46
24	0.09	2.16	0.18
25	0.12	3.00	0.02
26	0.15	3.90	0.05
27	0.13	3.51	0.33
28	0.10	2.80	0.67
29	0.10	2.90	1.29
30	0.05	1.50	1.05
	1.00	25.41	7.50

The Mean Value $\mu = \sum_{all\ x} xf(x)$

= 1.00 + 1.26 + 1.54 + 1.84 + 2.16 +
3.00 + 3.90 + 3.51 + 2.80 + 2.90 + 1.50
= 25.41

✓ Q1

The Standard Deviation $\sigma = \sqrt{\sum_{all\ x} (x - \mu)^2 f(x)}$

$= \sqrt{\sum_{all\ x} (x - 25.41)^2 f(x)}$

$= \sqrt{7.50} = 2.74$

Note:
1.46 + 1.17 + 0.81 + 0.46 + 0.18 + 0.02 + 0.05 + 0.33 + 0.67 + 1.29 + 1.05 = 7.50

(2) a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?

✓ Q2a

$P(25.41 - 2.74 \leq x \leq 25.41 + 2.74) =$
$P(22.67 \leq x \leq 28.15) =$
$P(23 \leq x \leq 28) = f(23) + f(24) + f(25) + f(26) + f(27) + f(28)$
$= 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10$
$= 0.67$ or 67%

b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?

✓ Q2b

$P(25.41 - 2*2.74 \leq x \leq 25.41 + 2*2.74) =$
$P(19.93 \leq x \leq 30.89) =$
$P(20 \leq x \leq 30) =$
$f(20) + f(21) + f(22) + f(23) + f(24) + f(25) + f(26) + f(27) + f(28) + f(29) + f(30) =$
1.00 or 100%

(3) Derive the distribution of the option value, ignoring interest and commissions.

Stock Price (x)	Probability f(x)	Option Value (v)
20	0.05	0
21	0.06	0
22	0.07	0
23	0.08	0
24	0.09	0
25	0.12	0
26	0.15	1
27	0.13	2

28	0.10	3
29	0.10	4
30	0.05	5
	1.00	

Thus, the following is the distribution of the option values, adding all probabilities where the option price is 0 together:

Option Price (v)	Option Probability g(v)
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05
	1.00

✓ Q3

(4) What is the probability that the option will have positive value?

$$\begin{aligned} P(v > 0) &= P(1) + P(2) + P(3) + P(4) + P(5) \\ &= 0.15 + 0.13 + 0.10 + 0.10 + 0.05 \\ &= 0.53 \text{ or } 53\% \end{aligned}$$

✓ Q4

OR

$P(v > 0) = 1 - P(v \leq 0) = 1 - P(0) = 1 - 0.47 = 0.53$ or 53%

(5) Compute the mean and standard deviation of the value of the option.

Option Price (v)	Option Probability g(v)	vg(v)	$(v-1.36)^2$ g(v)
0	0.47	0.00	0.87
1	0.15	0.15	0.02
2	0.13	0.26	0.05
3	0.10	0.30	0.27
4	0.10	0.40	0.70
5	0.05	0.25	0.66
	1.00	1.36	2.57

The Mean Value $\mu = \sum_{all\ v} vg(v)$

$= 0.00 + 0.15 + 0.26 + 0.30 + 0.40 + 0.25$

$= 1.36$

The Standard Deviation $\sigma = \sqrt{\sum_{all\ v} (v - \mu)^2 g(v)}$

$= \sqrt{\sum_{all\ v} (x - 1.36)^2 g(v)}$

✓ Q5

$= \sqrt{0.87 + 0.02 + 0.05 + 0.27 + 0.70 + 0.66}$

$= \sqrt{2.57}$

$= 1.60$